

Empirical Analysis of Filter Feature Selection Criteria on Financial Datasets

Bui Quoc Trung
trungbq@soict.hust.edu.vn
Hanoi University of Science and
Technology
Hanoi, Vietnam

Tran Van Tri
tri.tranvan9965@gmail.com
Hanoi University of Science and
Technology
Hanoi, Vietnam

Bui Thi Mai Anh*
anhbtm@soict.hust.edu.vn
Hanoi University of Science and
Technology
Hanoi, Vietnam

Abstract

High dimensionality is one of the data quality problems that affects the performance of machine learning models. Feature selection which aims to identify and remove as many redundant and irrelevant features as possible allows to boost the overall performance of the models while reducing the computational cost. However, the choice of an appropriate feature selection method is still a big challenge as there is no the best selection criterion that fits to all datasets. It is then essential to comparatively analyze the performance of feature selection criteria according to different characteristics of high-dimensional datasets, particularly large financial datasets where features are highly-correlated and redundant. In this paper, we explore nine different feature selection criteria which are typically categorized into two classes: (i) information theoretical based criteria and (ii) similarity based criteria over seven public financial datasets. To the best of our knowledge, no previous comprehensive empirical investigation has been carried out to demonstrate the positive effects of feature selection criteria on financial data. Experimental results indicate that the information theoretical-based methods suffer from a high computation time in case of high dimensional data (i.e., high number of features) while the similarity-based methods require significant computations to deal with high volume dataset (i.e., high number of samples).

Keywords: financial datasets, feature selection criteria, feature scoring

ACM Reference Format:

Bui Quoc Trung, Tran Van Tri, and Bui Thi Mai Anh. 2022. Empirical Analysis of Filter Feature Selection Criteria on Financial Datasets.

In *Proceedings of ACM Conference (Conference'17)*. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Data has increasingly become large in terms of both number of samples and number of attributes in many disciplines such as financial analysis [25], stock analysis [20], bioinformatic [5], customer relationship [19], etc. This trending may cause problems to machine learning algorithms regarding learning performance and computational cost. Indeed, high dimensional data typically contains irrelevant and redundant information which seriously reduce the accuracy of learning algorithms. Feature selection thus becomes a frequently-used method to tackle the problem of data dimensionality [7]. The main objective of feature selection is to pick up optimal features from the original set of features according to a certain evaluation criterion (i.e., selection criterion). Feature selection is considered as a pre-processing step to machine learning which allows to increase considerably the overall performance of learning tasks while achieving more efficient computation [15].

Recent studies have categorized feature selection algorithms into two main approaches: (i) filter methods and (ii) wrapper methods [15]. The filter approach relies on statistical analysis of features to evaluate their relevance without involving any learning algorithm. The selection process consists of two phases: (i) scoring each feature according to a given selection criterion (ii) filtering out low ranking score features. In contrast, the wrapper approach aims to search in the feature space a subset of features which allows to maximize the performance of a pre-defined learning algorithm. While the former is independent of all the learning models and tends to provide a generic selection of features, the latter has to perform iteratively the given algorithm to find optimal solution, leading to a more expensive computational performance. A trade-off between filter and wrapper approaches, embedded methods, is required to target to a specific learning algorithm while reducing computational resource [15]. Due to the computational efficiency, the filter approach is typically chosen to deal with high dimensional data, particularly when the number of feature is very large [3]. This work focuses on filter based methods and their application in the context of financial high dimensionality data.

*This author is the corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference'17, July 2017, Washington, DC, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM. . \$15.00

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

Large-scale datasets in terms of both size (i.e., number of samples) and dimension (i.e., number of features) may pose significant challenges to filter feature selection approaches. One of obvious challenges comes from the fact that results obtained from different filter algorithms on the same dataset may be discrepant. Based on the selection principles, there are various filtering algorithms which are mainly categorized into two families: (i) similarity-based and (ii) information theoretical-based criteria [15]. The first group exploits the ability of features to preserve data similarity to define their relevance. Fisher score [9], ReliefF [23], Laplacian score [10] etc. are instances of similarity-based methods. The second group measures the importance of features based on their correlation with class labels including Mutual Information Maximization (MIM) [14], Mutual Information Feature Selection (MIFS) [2], Minimum Redundancy Maximum Relevance (mRMR) [21], etc. These methods may act differently depending on the characteristics and the dimensionality of data. Regarding the impact of feature selection methods on the performance of classification task, recent studies have showed that the contradiction and inconsistency in feature subset selected by different approaches may be due to the high dimensionality of data in which there is a high degree of correlation between features [6, 11, 22]. Consequently, when the resulting feature subset is processed by the learning algorithm, the performance is also impacted, leading to the result discrepancy problem. It is then difficult to choose an appropriate feature selection method because there is no a perfect solution that fits to all datasets.

Feature selection approaches have emphasized their success in the research field of financial technology where the data is sparse, noisy and highly correlated [12, 16, 25]. Most studies focused on applying different feature selection methods to improve the performance of specific learning tasks. Liang et al. studied the effect of filter and wrapper approaches on financial distress prediction considering various learning models and found that there is no the best combination of the feature selection method and the classification technique [16]. A comparative study between three filter methods to solve the problem of credit scoring proposed by Trivedi [25] has showed that the combination of Random Forest and Chi Square has achieved the best performance. However, all of these works considered small-scale financial datasets (i.e., about 15 to 40 features and less than 1000 samples). We realize that there is a lack of empirical comparison on the impact of various feature selection approaches considering financial high dimensionality data. In this paper, we explore the impact of nine filter-based algorithms over seven public datasets in the context of the credit scoring task. The main contribution of our work is to analyze the pros and cons of different feature selection filter approaches when the data is high dimensional in terms of both volume and dimensionality. We compare these approaches regarding

the learning performance as well as computational cost and memory requirement.

The rest of this paper is structured as follows. In Section 2, we introduce the background of filter feature selection algorithms. Section 3 present empirical settings including the datasets, learning model and parameters. Section 4 focuses on the empirical analysis of filter algorithms over selected financial datasets in which we aim to address aforementioned research questions. Finally, Section 5 concludes with some future directions.

2 Background

In this section, we introduce traditional feature selection approaches which are widely used in the literature. We broadly categorize them into (i) similarity based and (ii) information theoretical based approaches.

2.1 Similarity-based Approaches

The methods in this family aim to evaluate the relevance of features by their ability to preserve the data similarity. Given a dataset of n instances and d features which can be described by a $n \times d$ dimensional matrix $\mathcal{X}$. The data similarity is calculated for each pairwise of instances and denoted as a $n \times n$ dimensional matrix $\mathcal{S}$. The selection of relevant features can be examined as an optimization problem in which we have to select the top k features by maximizing their utility in terms of data similarity (see Equation 1).

$$\max_{\hat{\mathcal{S}}} U(\hat{\mathcal{S}}) = \max_{\hat{\mathcal{S}}} \sum_{f \in \hat{\mathcal{S}}} U(f) \quad (1)$$

where $U(f)$ denotes the utility of feature f , $\hat{\mathcal{S}} \subset \mathcal{S}$ is the subset of k features. The different methods in this family vary by their mechanism to calculate the feature utility.

2.1.1 Laplacian score The calculation of the Laplacian score for each feature consists of three steps [10]. In the first step, the similarity matrix of $n \times n$ dimension S is defined where $s(i, j)$ denotes the similarity of each pair of instances x_i and x_j (Equation 2).

$$s(i, j) = e^{-\frac{(\|x_i - x_j\|)^2}{\sigma}} \quad (2)$$

where σ is a pre-defined constant, x_j is among the k nearest neighbours of x_i , otherwise $s(i, j) = 0$. In the second step, the diagonal matrix D is built such that $d(i, i) = \sum_{j=1}^n s(i, j)$. The Laplacian matrix L is defined as $L = D - S$. Finally, the Laplacian score of a feature f_i is computed as

$$\mathcal{L}(f_i) = \frac{\tilde{f}_i^T \cdot L \cdot \tilde{f}_i}{\tilde{f}_i^T \cdot D \cdot \tilde{f}_i} \text{ where } \tilde{f}_i = f_i - \frac{f_i^T \cdot D \cdot 1}{1^T \cdot D \cdot 1} \cdot 1 \quad (3)$$

The Laplacian score is calculated for each individual feature. The selection of k most relevant features is carried out by greedily picking the top k features with the smallest Laplacian scores.

2.1.2 SPEC While Laplacian score is only applied in unsupervised scenario, SPEC is one of its extensions which can be applied in both supervised and unsupervised scenarios. In the case of unsupervised scenario, the similarity $s(i, j)$ is measured by RBF kernel [8].

$$s(i, j) = e^{-\frac{\|x_i - x_j\|^2}{2\sigma^2}} \quad (4)$$

While in the supervised scenario, the label classes will be used to calculate the similarity between two instances as in Equation 5.

$$s(i, j) = \begin{cases} \frac{1}{n_l} & \text{if } y_i = y_j = l \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where n_l is the number of data samples in the l^{th} class. The similarity matrix S and the diagonal matrix D are computed. The Laplacian matrix is then normalized $L_{norm} = D^{-\frac{1}{2}} \cdot (D - S) \cdot D^{-\frac{1}{2}}$. The feature relevance is measured by computing the SPEC score as in Equation 6.

$$SPEC(f_i) = \frac{\sum_{j=2}^n \alpha_j^2 \gamma(\lambda_j)}{\sum_{j=2}^n \alpha_j^2} \quad (6)$$

where $\alpha_i = \cos(v_i, f_i)$, λ_j and v_j denote the j^{th} eigenpair of the normalized Laplacian matrix L_{norm} . $\gamma(\cdot)$ is an increasing function to reduce noise. In the case that the noise of data is free, $\gamma(x) = x$. The top k features with the highest SPEC scores will be selected.

2.1.3 Fisher score Fisher score is a supervised feature selection algorithm [9]. The features are selected to assure that the feature values of samples within the same class are similar while the feature values of samples from different classes are dissimilar. The Fisher score of a feature f_i is calculated as follows.

$$Fisher(f_i) = \frac{\sum_{j=1}^c n_j (\mu_{ij} - \mu_i)^2}{\sum_{j=1}^c n_j \sigma_{ij}^2} \quad (7)$$

where n_j is the number of instances in the class j , μ_i is the mean value of the feature f_i while μ_{ij} and σ_{ij}^2 are the mean and variance values of f_i for all instances in the class j . The top k features with the highest Fisher scores will be selected.

2.1.4 ReliefF The main idea of ReliefF is to select features by separating instances from different classes [23]. Given an instance x_j , let W_j be the set of nearest within-class instances of x_j and B_{jy} be the set of nearest instance of x_j in the class y . Given l samples randomly picked up from the original dataset with c class labels, the ReliefF score of a feature f_i is defined as follows.

$$\begin{aligned} ReliefF(f_i) &= \frac{1}{c} \sum_{j=1}^l \left(-\frac{1}{|W_j|} \sum_{x_r \in W_j} d(X(j, i) - X(r, i)) \right. \\ &\quad \left. + \sum_{y \neq j} \frac{1}{|B_{jy}|} \frac{p(y)}{1-p(y)} \sum_{x_r \in B_{jy}} d(X(j, i) - X(r, i)) \right) \end{aligned} \quad (8)$$

where $|W_j|$ and $|B_{jy}|$ are the size of W_j and B_{jy} , respectively. $p(y)$ is the ratio of instances in class y .

2.2 Information Theoretical-based Approaches

Information Theoretical based assess the relevance of features by applying heuristic criteria. The key concept of information theory is the *entropy* which measures the uncertainty of a discrete random variable [8].

$$H(X) = - \sum_{x_i \in X} P(x_i) \log(P(x_i)) \quad (9)$$

where $X = \{x_i\}_{i=1}^n$ is the random discrete variable, $P(x_i)$ denotes the probability of x_i over all possible values of X .

The second concept, *conditional entropy*, represents the uncertainty of the variable X given another random discrete variable Y (Equation 10).

$$H(X|Y) = - \sum_{y_i \in Y} \sum_{x_i \in X} P(x_i|y_i) \log(P(x_i|y_i)) \quad (10)$$

where $P(x_i|y_i)$ is the conditional probability of x_i given y_i .

The information gain (a.k.a. mutual information) between two variables X and Y allows to measure the amount of information shared by these variables.

$$I(X; Y) = H(X) - H(X|Y) = \sum_{x_i \in X} \sum_{y_i \in Y} P(x_i, y_i) \log \frac{P(x_i, y_i)}{P(x_i)P(y_i)} \quad (11)$$

The subset $\hat{S}$ of k most relevant features is built by adding sequentially a new feature F_i with the highest relevance score to $\hat{S}$. In the rest of this section, we introduce representative selection criteria in this family. Each criterion shows its mechanism to calculate the feature relevance score based on information theory.

2.2.1 Mutual Information Maximization (MIM) Under the assumption that a feature which has a strong correlation with the class label allows to achieve good classification performance, the MIM criterion aims to select k most relevant features with the highest mutual information scores [14] (Equation 12).

$$MIM(F_i) = I(F_i; Y) \quad (12)$$

where $I(F_i; Y)$ denotes the mutual information between the feature F_i and the class variable Y .

2.2.2 Mutual Information Feature Selection (MIFS) The MIM criterion considers individually features however a good feature is not only highly correlated with the class label but also should be as independent as possible to other features. The MIFS criterion takes into account both the feature relevance and feature redundancy to select k most relevant features [2] (Equation 13).

$$MIFS(F_i) = I(F_i; Y) - \beta \sum_{F_j \in \hat{S}, i \neq j} I(F_i; F_j) \quad (13)$$

where $I(F_i; Y)$ denotes the relevance of feature F_i to the class label Y while the second term represents the total correlation between this feature and the other selected features. β is the penalization parameter to emphasize the importance of the redundancy.

2.2.3 Minimum Redundancy Maximum Relevance (MRMR) MRMR is a variant of MIFS proposed by Peng et al. [21] to fix the value of the parameter β as the reverse of the number of selected features $\frac{1}{|\hat{S}|}$ (Equation 14).

$$MRMR(F_i) = I(F_i; Y) - \frac{1}{|\hat{S}|} \sum_{F_j \in \hat{S}} I(F_i; F_j) \quad (14)$$

2.2.4 Conditional Infomax Feature Extraction (CIFE) The CIFE criterion aims to not only minimize the feature redundancy but also maximize the conditional redundancy between the under-consideration feature and the selected features given the class label [17].

$$CIFE(F_i) = I(F_i; Y) - \sum_{F_j \in \hat{S}} I(F_i; F_j) + \sum_{F_j \in \hat{S}} I(F_j; F_i|Y) \quad (15)$$

where the third terms introduce the conditional redundancy among F_i and the selected features in $\hat{S}$ given the class label Y which is calculated by Equation 16).

$$I(F_j; F_i|Y) = H(F_j|Y) - H(F_j|F_i, Y) \quad (16)$$

2.2.5 Joint Mutual Information (JMI) JMI is a variant of CIFE proposed by Brown et al. in which they added adaptive parameters to the redundancy and conditional redundancy [4].

$$JMI(F_i) = I(F_i; Y) - \frac{1}{|\hat{S}|} \sum_{F_j \in \hat{S}} I(F_i; F_j) + \frac{1}{|\hat{S}|} \sum_{F_j \in \hat{S}} I(F_j; F_i|Y) \quad (17)$$

While the similarity-based methods evaluate features individually to maximize the data similarity, thus fail to tackle feature redundancy, the information theoretical based approaches take advantages of both feature relevance and feature redundancy. However, the former can be applied to both supervised and unsupervised classification. The latter, in contrast, only works with the supervised scenario as the relevance of a feature is measured with the guide of class labels. In the next section, we apply these feature selection methods to tackle the problem of high-dimensional data in order to provide an insight of the pros and cons of these approaches.

3 Experimental Design

3.1 Credit Scoring Model

Credit scoring is a statistical modeling technique which aims to measure the risk for a bank to grant a loan to credit applicants or credit accounts [1]. The decision on whether or not to approve a loan is made according to the value of the

credit score which is typically calculated through a variety of features such as consumer behaviors (e.g., amount, interest rate on loans etc.) and credit profile (e.g., age, marital status, monthly income, personal savings, etc.) [24]. From the practical perspective, credit scoring task can be solved by utilizing classification models in which credit applicants are categorized into different groups according to their important score [18, 25]. The goal of our study is not to investigate the performance of different classifiers, we therefore choose a widely used supervised learning algorithm, Support Vector Machine [18], to tackle the problem of credit scoring. Feature selection techniques are separately applied on the same dataset and using the same model configuration.

The financial data is first pre-processed using various filter feature selection methods (see Section 2) to reduce dimension and exclude irrelevant features. This phase consists of two steps:

- A given filter selection algorithm assigns to each feature a relevance score and produces a ranking list ranged from the highest to the lowest relevancy.
- Low ranking score features are filtered out according to a pre-defined threshold (i.e., a cut-off value)

The training and testing phases are then carried out with the result data of the pre-processing phase. We split the data into 10 equally sized folds with random sampling. The model will be trained on fold i and test on fold $i + 1$. The final performance of the learning model is the average of all testing folds.

Table 1. Benchmark Datasets

Dataset	Number of Samples	Number of Features
CreditScore	80000	304
FintechUser	27000	29
CreditCard	284807	29
HomeCredit	307516	304
CreditRisk	855000	71
FinancialRisk	1048575	45
VehicleLoan	233154	39

3.2 Datasets

As aforementioned, our work aims to deal with financial high dimensional data. We conduct an empirical study on seven public datasets* [13] in the domain of finance technology. Table 1 shows the overall statistical information of these datasets. It can be observable that the number of features is also varied, ranging from 29 to 304 features while the number of observation is from 80.000 to more than one million. CreditScore and HomeCredit have the highest number of features, 304 features. FinancialRisk is characterized by only

*<https://github.com/JLZml/Credit-Scoring-Data-Sets>

Table 2. Performance Comparison on 7 datasets

Dataset	Performance	Feature Selection Approaches								
		Lap.	SPEC	Fisher	ReliefF	MIM	MIFS	MRMR	CIFE	JMI
CreditScore	F1-Score	0.59	0.45	0.72	0.72	0.72	0.54	0.61	0.66	0.73
	ACC	0.9	0.88	0.93	0.93	0.93	0.89	0.9	0.91	0.93
	AUC	0.75	0.66	0.82	0.81	0.81	0.71	0.76	0.78	0.83
FintechUser	F1-Score	0.54	0.01	0.54	0.54	0.54	0.48	0.47	0.51	0.54
	ACC	0.74	0.68	0.75	0.75	0.74	0.72	0.72	0.73	0.75
	AUC	0.67	0.5	0.67	0.67	0.67	0.64	0.63	0.65	0.67
CreditCard	F1-Score	0.4	0.4	1	0.86	0.67	0.67	0.62	0.33	0.8
	ACC	1	1	1	1	1	1	1	1	1
	AUC	0.62	0.62	1	0.88	0.75	0.75	0.74	0.62	0.85
HomeCredit	F1-Score	0.01	0.03	0.06	0.01	0.05	0.02	0.02	0.02	0.05
	ACC	0.92	0.91	0.92	0.92	0.92	0.91	0.91	0.91	0.92
	AUC	0.5	0.51	0.51	0.5	0.51	0.5	0.5	0.5	0.51
CreditRisk	F1-Score	0.95	0.01	0.98	0.98	0.98	0.7	0.71	0.78	0.98
	ACC	0.99	0.94	1	1	1	0.97	0.78	0.98	1
	AUC	0.97	0.5	0.98	0.99	0.98	0.77	0.97	0.82	0.99
FinancialRisk	F1-Score	0.04	0.04	0.18	0.15	0.11	0.05	0.02	0.12	0.14
	ACC	0.8	0.8	0.81	0.8	0.81	0.8	0.8	0.8	0.81
	AUC	0.51	0.51	0.54	0.53	0.52	0.51	0.5	0.52	0.53
VehicleLoan	F1-Score	0.01	0.01	0.5	0.47	0.47	0.01	0.01	0.48	0.48
	ACC	0.77	0.77	0.85	0.84	0.84	0.78	0.77	0.84	0.84
	AUC	0.5	0.5	0.67	0.65	0.65	0.5	0.5	0.66	0.66

45 features but contains the highest number of samples, i.e., over one million samples.

3.3 Evaluation Metrics and Parameter Settings

In order to assess the feature selection mechanism, we use three metrics which are widely adopted to measure the performance credit scoring models including Accuracy, F1-Score and AUC [13].

3.3.1 Accuracy The accuracy of a classification model is the ratio of all the accurately classified samples and the total samples of the dataset. This measure can be calculated through the confusion matrix which represents four possible outcomes of a classifier to make prediction including *true positive* (TP), *true negative* (TN), *false positive* (FP) and *false negative* (FN) (Equation 18).

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

3.3.2 F1 Score F1 score is the most commonly used derivative of the *F-measure* and is defined by Equation 19

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (19)$$

This measure represents a balance between *precision* and *recall* of the classification model. F1-score is a better indicator than Accuracy particularly when the data is unevenly distributed.

3.3.3 AUC The final measure, AUC (Area Under the Curve) calculates the area under an ROC curve which describes relative trade-offs between the *true positive rate* (a.k.a TPR which is $TP/(TP + FN)$) and the *false positive rate* (a.k.a FPR which is $FP/(FP + TN)$).

Furthermore, computational time (in seconds) is also computed in this study as supplementary evaluation metric to compare the effectiveness of feature selection approaches.

4 Results

4.1 Evaluation Plan

In this section, we present a comparative empirical study to examine the impact of two families of feature selection algorithms on the learning performance and computational cost when the data is high dimensional. The same experimental procedure (as described in Section 3.1) has been applied on seven datasets. The number of selected features was empirically fixed as 25%. We first examine the learning performance of the credit scoring model considering the similarity-based

and information theoretical based approaches. We then evaluate the computational time of each feature selection method on different datasets to explore the impact of both volume and dimensionality.

4.2 Empirical Results

We present our experimental results through the following research questions.

4.2.1 RQ1: Which feature selection approach is capable to deal with high dimensional data in terms of learning performance? Table 2 shows the performance of the credit scoring model on seven datasets for all feature selection approaches.

First, regarding two datasets with the highest number of features, CreditScore and HomeCredit (304 features), the results are more or less the same for Fisher, MIM and JMI approaches. For example, with respect to CreditScore the model achieved the best performance with F1-Score from 0.72 to 0.73 points, AUC from 0.81 to 0.83 points when utilizing Fisher, ReliefF, MIM and JMI. The accuracy of the model was 93% for all those approaches. Considering HomeCredit, it seems that the model did not work well on this dataset due to the fact that the value of AUC approximated to 0.5 point. The methods Fisher, MIM and JMI however still allow to obtain the highest F1-Score's value (around 0.05 point). It is important to note that the value of accuracy (ACC) was very high on this dataset (around 0.91 to 0.92 points). The high degree of imbalance within the dataset (i.e., more than 90% of the samples are labelled as 0 while about 10% of them are labelled as 1) is the main reason of this phenomena. In order to improve the performance of classification model, it should consider solutions to deal with highly skewed distribution data.

Second, considering the remaining datasets, it can be observed that with three approaches Fisher, ReliefF and JMI, the model has achieved the best performance for all evaluation metrics. Furthermore, the Laplacian method has showed promising results on two datasets FintechUser and CreditRisk, as the value of AUC is comparable to other approaches. The case of FinancialRisk is quite similar to HomeCredit. Because of the highly imbalanced data (about 95% as majority class), the classification performance is not good.

Finally, the testing results on AUC for all the datasets have figured out that the method SPEC was not efficient. Indeed, the value of AUC was 0.62 and 0.66 for CreditScore and CreditCard, respectively while it was around 0.5 for the other datasets.

4.2.2 RQ2: How do the massive volume and high dimensionality affect the feature selection approaches?

In order to answer to this research question, we have conducted experiments to review the resource consumption of all the feature selection approaches.

Table 3. Comparison on Computational Time of Similarity-based Approaches considering 20.000 samples for each dataset

Dataset	Dim.	Feature Selection Approaches			
		Lap.	Fisher	SPEC	ReliefF
CreditScore	304	35.6	79.8	393	425
HomeCredit	304	27.8	74.3	506	283
CreditRisk	71	25.4	69.3	420	300
FinancialRisk	45	22.5	53.9	369.4	255.5
VehicleLoan	39	20.2	51.7	345.7	271.6
FintechUser	29	18.1	63.1	330.6	410
CreditCard	29	17.6	63.9	312.6	278

Observation 1. We firstly have compared the computational time of the similarity-based approaches. As all the approaches of this family have to calculate an $n \times n$ dimensional matrix (with n is the number of instances of the dataset), they should depend on the volume of dataset. For the purpose of observing the impact of data dimensionality, we have fixed the number of instances for all datasets as 20.000. Table 3 shows the time taken (in seconds) of each algorithm to perform the feature selection with respect to seven datasets. For each algorithm, it can be noted that the time consumption is more or less the same without regard for the dimensionality of dataset. For example, considering the Fisher approach, the time taken to select the most salient features is about 79.8 seconds on the CreditScore dataset which has 304 features while it is 69.3 seconds regarding CreditRisk with 71 features. Additionally, the results have figured out that Laplacian is the most efficient while SPEC and ReliefF are less competent in terms of computational time.

Table 4. Comparison on Computational Time of Similarity-based Approaches considering the whole datasets

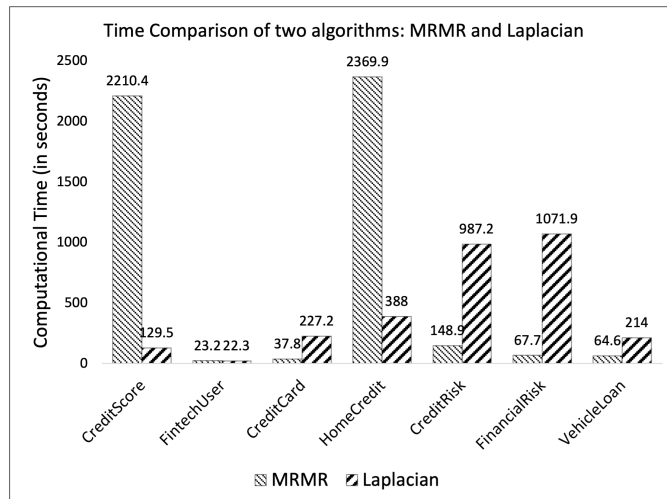
Dataset	Feature Selection Approaches			
	Lap.	Fisher	SPEC	ReliefF
CreditScore	129.4	290.2	1429	850
HomeCredit	388	1036.8	7061	2172
CreditRisk	987	2693.3	16322.7	6412.5
FinancialRisk	1071.8	2567.6	17597	6694.1
VehicleLoan	213.9	547.6	3661.3	1582.1
FintechUser	22.2	77.4	405.7	556.4
CreditCard	227.2	825	4035.4	1973.8

Observation 2. Table 4 shows the time taken of each algorithm to select relevant features on the whole volume of seven datasets. It is reasonable that the computational time significantly increases when the number of instances is extremely high. For example, regarding the FinancialRisk dataset, while the Laplacian algorithm took only 22.5 seconds to process 20.000 samples, it has required more than 47 times

Table 5. Comparison on Computational Time of Information Theoretical-based Approaches considering the whole datasets

Dataset	Dim.	Volume	Feature Selection Approaches				
			MIM	MIFS	MRMR	CIFE	JMI
CreditScore	304	80,000	2305.6	2342.6	2210.4	2231.7	2350.4
HomeCredit	304	307,516	2506.3	2563.7	2369.9	2370.7	2580.8
CreditRisk	71	855,000	162.1	158.3	148.9	153.3	154.9
FinancialRisk	45	1,048,575	68.9	67.9	67.7	68.9	62.5
VehicleLoan	39	233,154	70.2	67.4	64.6	64.1	69.7
FintechUser	29	27,000	37.9	35.9	23.7	23.4	36.8
CreditCard	29	284,807	46.3	46.7	37.8	35.8	38.8

to process one million samples. Laplacian is consistently found the most efficient algorithm comparing to the others in the similarity-based family. Finally, it is important to note that all these methods require ultrahigh memory resource requirement, particularly when considering massive volume datasets.

**Figure 1.** Comparison on Computational Time of two algorithms, MRMR and Laplacian

Observation 3. We have performed the same experiment to compare the computational time of the information theoretical based approaches. While the similarity-based approaches consider each feature individually and compute the similarity matrix for all instances, the information theoretical based methods calculate the correlation between features. The number of features then strongly affects the computational time of those approaches. As can be seen in Table 5, the time consumption is very high for all five algorithms on two datasets, CreditScore and HomeCredit due to their highest dimensionality (304 features). Furthermore, the number of samples of the dataset slightly affect the computational time of information theoretical-based approaches. Indeed, regarding the JMI method, it took 36.8 and 38.8 seconds to

select relevant features from FintechUser with 27,000 samples and Credit with more than 200,000 samples, respectively. Finally, the empirical results have showed that MRMR is the most efficient approach comparing to the others.

Observation 4. Figure 1 shows the comparison of two most efficient approaches, Laplacian from the similarity-based family and MRMR from the information theoretical based family. It is observable that MRMR is better than Laplacian in case of small-scale volume datasets. For example, the Laplacian method spent 227.2 seconds, six times as much as MRMR on the CreditCard dataset. In case of massive volume such as FinancialRisk, the time consumption of Laplacian is 15 times as much as MRMR. However, Laplacian outperforms MRMR in the circumstance of high dimensional data. Indeed, MRMR took 2210.4 and 2369.9 seconds while Laplacian spent only 129.5 and 388 seconds to perform the feature selection process on CreditScore and HomeCredit, respectively.

5 Conclusion

Research in feature selection is ongoing and continue to expand as real applications become more and more extensive and automatic data collection techniques are increasingly required leading to high-correlated, redundant and multi-dimensional data. This paper has empirically analyzed the effect of a wide range of filter feature selection approaches in terms of both performance learning and computational cost targeted to financial high-dimensional datasets. We find out that the information theoretical-based methods suffer from a high computation time in case of high dimensional data (i.e., high number of features) while the similarity-based methods require significant computations to deal with high volume dataset (i.e., high number of samples). The empirical results figure out that Laplacian and MRMR achieve the best performance in terms of computational time while Fisher, ReliefF and JMI methods allow to gain clearly distinguishable learning performance from the other approaches.

References

- [1] Raymond Anderson. 2007. *The credit scoring toolkit: theory and practice for retail credit risk management and decision automation*. Oxford University Press.

- [2] Roberto Battiti. 1994. Using mutual information for selecting features in supervised neural net learning. *IEEE Transactions on neural networks* 5, 4 (1994), 537–550.
- [3] Verónica Bolón-Canedo, Noelia Sánchez-Marroño, and Amparo Alonso-Betanzos. 2015. Recent advances and emerging challenges of feature selection in the context of big data. *Knowledge-based systems* 86 (2015), 33–45.
- [4] Gavin Brown, Adam Pocock, Ming-Jie Zhao, and Mikel Luján. 2012. Conditional likelihood maximisation: a unifying framework for information theoretic feature selection. *The journal of machine learning research* 13 (2012), 27–66.
- [5] Zheng Chen, Meng Pang, Zixin Zhao, Shuainan Li, Rui Miao, Yifan Zhang, Xiaoyue Feng, Xin Feng, Yexian Zhang, Meiyu Duan, et al. 2020. Feature selection may improve deep neural networks for the bioinformatics problems. *Bioinformatics* 36, 5 (2020), 1542–1552.
- [6] Baljinder Ghotra, Shane McIntosh, and Ahmed E Hassan. 2017. A large-scale study of the impact of feature selection techniques on defect classification models. In *2017 IEEE/ACM 14th International Conference on Mining Software Repositories (MSR)*. IEEE, 146–157.
- [7] D Asir Antony Gnana, S Appavu Alias Balamurugan, and E Jebamalar Leavline. 2016. Literature review on feature selection methods for high-dimensional data. *International Journal of Computer Applications* 136, 1 (2016), 9–17.
- [8] Robert M Gray. 2011. *Entropy and information theory*. Springer Science & Business Media.
- [9] Peter E Hart, David G Stork, and Richard O Duda. 2000. *Pattern classification*. Wiley Hoboken.
- [10] Xiaofei He, Deng Cai, and Partha Niyogi. 2005. Laplacian score for feature selection. *Advances in neural information processing systems* 18 (2005).
- [11] Firuz Kamalov and Fadi Thabtah. 2017. A feature selection method based on ranked vector scores of features for classification. *Annals of Data Science* 4, 4 (2017), 483–502.
- [12] Nikita Kozodoi, Stefan Lessmann, Konstantinos Papakonstantinou, Yiannis Gatsoulis, and Bart Baesens. 2019. A multi-objective approach for profit-driven feature selection in credit scoring. *Decision support systems* 120 (2019), 106–117.
- [13] Stefan Lessmann, Bart Baesens, Hsin-Vonn Seow, and Lyn C Thomas. 2015. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research* 247, 1 (2015), 124–136.
- [14] David D Lewis. 1992. Feature selection and feature extraction for text categorization. In *Speech and Natural Language: Proceedings of a Workshop Held at Harriman, New York, February 23-26, 1992*.
- [15] Jundong Li, Kewei Cheng, Suhang Wang, Fred Morstatter, Robert P Trevino, Jiliang Tang, and Huan Liu. 2017. Feature selection: A data perspective. *ACM computing surveys (CSUR)* 50, 6 (2017), 1–45.
- [16] Hongliang Liang, Lu Sun, Meilin Wang, and Yuxing Yang. 2019. Deep learning with customized abstract syntax tree for bug localization. *IEEE Access* 7 (2019), 116309–116320.
- [17] Dahua Lin and Xiaoou Tang. 2006. Conditional infomax learning: An integrated framework for feature extraction and fusion. In *European conference on computer vision*. Springer, 68–82.
- [18] Sebastián Maldonado, Juan Pérez, and Cristián Bravo. 2017. Cost-based feature selection for support vector machines: An application in credit scoring. *European Journal of Operational Research* 261, 2 (2017), 656–665.
- [19] David Alfred Ostrowski. 2014. Feature selection for twitter classification. In *2014 IEEE International Conference on Semantic Computing*. IEEE, 267–272.
- [20] Kunal Pahwa and Neha Agarwal. 2019. Stock market analysis using supervised machine learning. In *2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon)*. IEEE, 197–200.
- [21] Hanchuan Peng, Fuhui Long, and Chris Ding. 2005. Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Transactions on pattern analysis and machine intelligence* 27, 8 (2005), 1226–1238.
- [22] Khairan D Rajab. 2017. New hybrid features selection method: A case study on websites phishing. *Security and Communication Networks* 2017 (2017).
- [23] Marko Robnik-Šikonja and Igor Kononenko. 2003. Theoretical and empirical analysis of ReliefF and RReliefF. *Machine learning* 53, 1 (2003), 23–69.
- [24] Lyn Thomas, Jonathan Crook, and David Edelman. 2017. *Credit scoring and its applications*. SIAM.
- [25] Shrawan Kumar Trivedi. 2020. A study on credit scoring modeling with different feature selection and machine learning approaches. *Technology in Society* 63 (2020), 101413.